

Quiz 1.4 – Dimensional Analysis Quiz

Name: Key

Question 1

Convert the value $80.0 \frac{\text{miles}}{\text{h}}$ to units of m/s

$$\frac{80.0 \text{ miles}}{1 \text{ hour}} \times \frac{1.609 \text{ km}}{1 \text{ mile}} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ hour}}{60 \text{ min}} \times \frac{1 \text{ min}}{60 \text{ s}} = 35.7556 \text{ m/s} \rightarrow 35.8 \text{ m/s}$$

Question 2

Light travels at a speed of $2.998 \times 10^8 \text{ m/s}$

- How many s does it take for light to travel from the surface of the earth to the moon and back (478,000 miles)?

$$\frac{478,000 \text{ miles}}{1 \text{ mile}} \times \frac{1.609 \text{ km}}{1 \text{ km}} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ s}}{2.998 \times 10^8 \text{ m}} = 2.56538 \text{ s} \rightarrow 2.57 \text{ s}$$

- How far does light travel in one minute?

$$\frac{1 \text{ min}}{1 \text{ min}} \times \frac{60 \text{ s}}{1 \text{ s}} \times \frac{2.998 \times 10^8 \text{ m}}{1 \text{ s}} = 1.7988 \times 10^{10} \text{ m} \rightarrow 1.799 \times 10^{10} \text{ m}$$

Question 3

Convert the following temperatures from K to $^{\circ}C$ or from $^{\circ}C$ to K 25.0 $^{\circ}C$ 376.5 K -12.3 $^{\circ}C$ 184.7 K 298.2 K 103.7 $^{\circ}C$ 260.8 K -88.7 $^{\circ}C$

Problem 4

Give the answer to the correct number of significant figures

- A person runs 1.00 miles in 7.46 minutes. Give her speed in $\frac{\text{m}}{\text{s}}$

$$\frac{1.00 \text{ miles}}{7.46 \text{ min}} \times \frac{1 \text{ min}}{60 \text{ s}} \times \frac{1.609 \text{ km}}{1 \text{ mile}} \times \frac{1000 \text{ m}}{1 \text{ km}} = 3.5947 \text{ m/s} \rightarrow 3.59 \text{ m/s}$$

- A metal sample weighs 10.78 kg and has $V = 0.954 \text{ L}$. Give the density in $\frac{\text{g}}{\text{ml}}$

$$\frac{10.78 \text{ kg}}{0.954 \text{ L}} \times \frac{1000 \text{ g}}{1 \text{ kg}} \times \frac{1 \text{ L}}{1000 \text{ ml}} = 11.2998 \text{ g/ml} \rightarrow 11.3 \text{ g/ml}$$

Problem 5

Based on Table 1.4 in your textbook, guess which metal is described in the problem above

Lead